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Section: F

Topic: LAB 7 - Fuzzing for Input Validation Bugs (Screenshots)

Step 2:

```
PS C:\college_files\se lab\lab 7> python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north
Traceback (most recent call last):
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 17, in main
    result = parse_zone(user_input)
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 2, in parse_zone
    name, area = text.split(':')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

```
PS C:\college_files\se lab\lab 7> python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

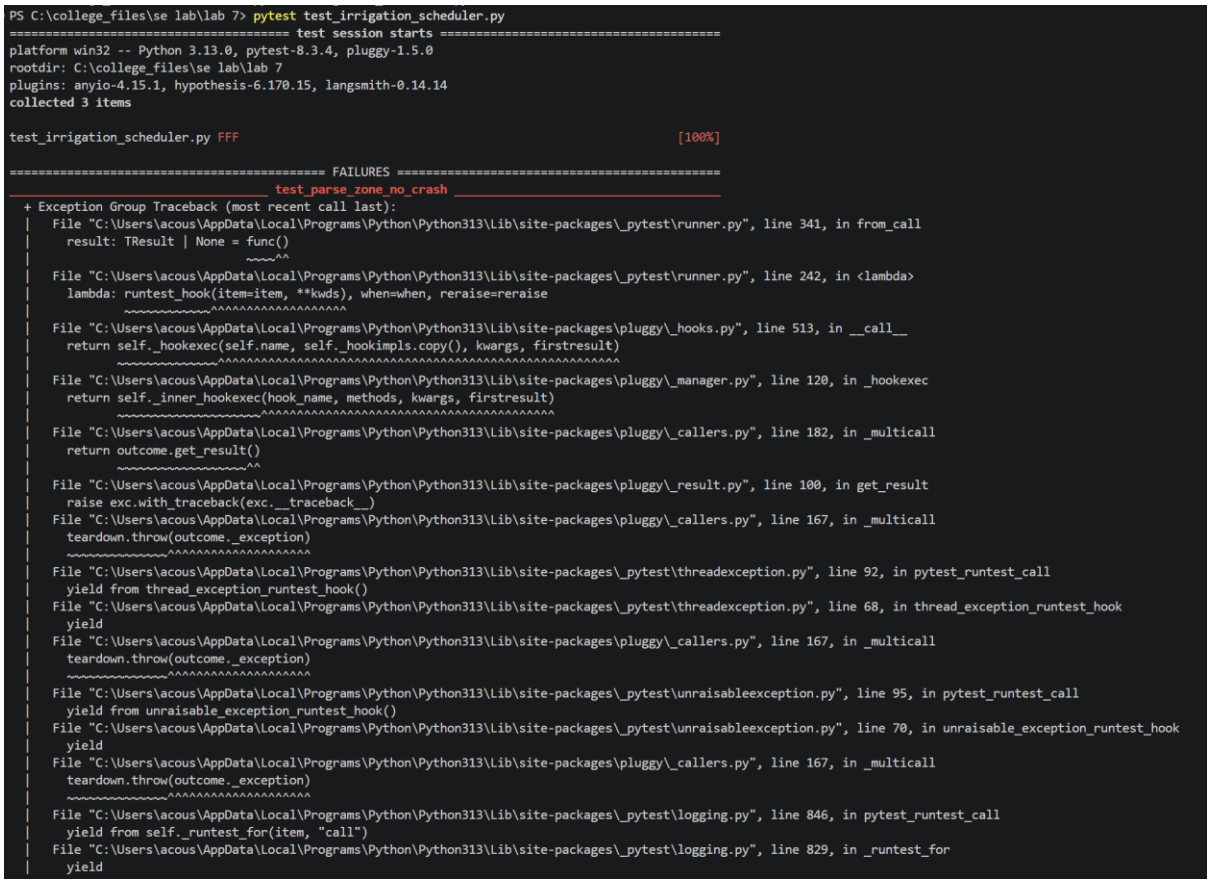
==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,,50
Traceback (most recent call last):
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 22, in main
    result = moisture_average(user_input)
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 6, in moisture_average
    values = [int(r) for r in readings_csv.split(',')]
              ~~~^^^
ValueError: invalid literal for int() with base 10: ''
```

```
PS C:\college_files\se lab\lab 7> python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north:120
Zone and Area: ('North', 120.0)

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
Average Moisture: 41

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 120
Traceback (most recent call last):
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 31, in <module>
    main()
    ~~~~^^
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 27, in main
    result = water_needed(user_input)
  File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 10, in water_needed
    plot, rain = text.split('|')
    ^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4:



```
PS C:\college_files\se lab\lab 7> pytest test_irrigation_scheduler.py
========================================= test session starts =========================================
platform win32 -- Python 3.13.0, pytest-8.3.4, pluggy-1.5.0
rootdir: C:\college_files\se lab\lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.14
collected 3 items

test_irrigation_scheduler.py FFF                                                              [100%]

============================================== FAILURES ===============================================
_____________________________ test_parse_zone_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ~~~~~~~~~~~~~~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
```

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  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |     ~~~~~~~~~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 6, in test_parse_zone_no_crash
  |     def test_parse_zone_no_crash(s):
  |                                  ^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 8, in test_parse_zone_no_crash
    |     parse_zone(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 3, in parse_zone
    |     return name.strip().title(), float(area)
    |                                  ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
```

```
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 8, in test_parse_zone_no_crash
    |     parse_zone(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 3, in parse_zone
    |     return name.strip().title(), float(area)
    |                                  ~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 10, in test_parse_zone_no_crash
    |     assert False, f"parse_zone crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_zone crashed with: ValueError("could not convert string to float: ''") for input ':'
    | assert False
    | Failing test case: test_parse_zone_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 8, in test_parse_zone_no_crash
    |     parse_zone(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 2, in parse_zone
    |     name, area = text.split(':')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 10, in test_parse_zone_no_crash
    |     assert False, f"parse_zone crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_zone crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
    | assert False
    | Failing test case: test_parse_zone_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 8, in test_parse_zone_no_crash
    |     parse_zone(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 2, in parse_zone
    |     name, area = text.split(':')
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
```

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 10, in test_parse_zone_no_crash
    |     assert False, f"parse_zone crashed with: {e!r} for input {s!r}"
    | AssertionError: parse_zone crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_parse_zone_no_crash(
    |     s=None,  # or any other generated value
    | )
    +---------------------------------
_____________________________ test_moisture_average_no_crash _____________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                             ~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ~~~~~~~~~~~~~~~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
```

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|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
|     teardown.throw(outcome._exception)
|     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
|     return (yield)
|             ^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
|     teardown.throw(outcome._exception)
|     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
|     return (yield)
|             ^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
|     res = hook_impl.function(*args)
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
|     item.runtest()
|     ~~~~~~~~~~~~^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 1627, in runtest
|     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
|     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
|     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
|            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
|     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
|            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
|     raise exception.with_traceback(exception.__traceback__)
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
|     res = hook_impl.function(*args)
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
|     result = testfunction(**testargs)
|   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 14, in test_moisture_average_no_crash
|     def test_moisture_average_no_crash(s):
|                ^^^
|   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
|     raise the_error_hypothesis_found
| ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
+-+---------------- 1 ----------------
  | Traceback (most recent call last):
  |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 16, in test_moisture_average_no_crash
  |     moisture_average(s)
  |     ~~~~~~~~~~~~~~~~^^^
  |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 6, in moisture_average
  |     values = [int(r) for r in readings_csv.split(',')]
  |               ~~~^^^
  | ValueError: invalid literal for int() with base 10: ''
  |
  | During handling of the above exception, another exception occurred:
  |
```

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    +-+---------------- 1 ----------------
      | Traceback (most recent call last):
      |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 16, in test_moisture_average_no_crash
      |     moisture_average(s)
      |     ~~~~~~~~~~~~~~~~^^^
      |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 6, in moisture_average
      |     values = [int(r) for r in readings_csv.split(',')]
      |               ~~~^^^
      | ValueError: invalid literal for int() with base 10: ''
      |
      | During handling of the above exception, another exception occurred:
      |
      | Traceback (most recent call last):
      |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 18, in test_moisture_average_no_crash
      |     assert False, f"moisture_average crashed with: {e!r} for input {s!r}"
      | AssertionError: moisture_average crashed with: ValueError("invalid literal for int() with base 10: ''") for input ''
      | assert False
      | Failing test case: test_moisture_average_no_crash(
      |     s='',  # or any other generated value
      | )
      +---------------- 2 ----------------
      | Traceback (most recent call last):
      |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 16, in test_moisture_average_no_crash
      |     moisture_average(s)
      |     ~~~~~~~~~~~~~~~~^^^
      |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 6, in moisture_average
      |     values = [int(r) for r in readings_csv.split(',')]
      |                              ^^^^^^^^^^^^^^^^^^
      | AttributeError: 'NoneType' object has no attribute 'split'
      |
      | During handling of the above exception, another exception occurred:
      |
      | Traceback (most recent call last):
      |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 18, in test_moisture_average_no_crash
      |     assert False, f"moisture_average crashed with: {e!r} for input {s!r}"
      | AssertionError: moisture_average crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
      | assert False
      | Failing test case: test_moisture_average_no_crash(
      |     s=None,  # or any other generated value
      | )
      +-----------------------------------
```

```
_____ test_water_needed_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 341, in from_call
  |     result: TResult | None = func()
  |                              ~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 242, in <lambda>
  |     lambda: runtest_hook(item=item, **kwds), when=when, reraise=reraise
  |             ~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 182, in _multicall
  |     return outcome.get_result()
  |            ~~~~~~~~~~~~~~~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_result.py", line 100, in get_result
  |     raise exc.with_traceback(exc.__traceback__)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 92, in pytest_runtest_call
  |     yield from thread_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\threadexception.py", line 68, in thread_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 95, in pytest_runtest_call
  |     yield from unraisable_exception_runtest_hook()
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\unraisableexception.py", line 70, in unraisable_exception_runtest_hook
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 846, in pytest_runtest_call
  |     yield from self._runtest_for(item, "call")
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\logging.py", line 829, in _runtest_for
  |     yield
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\capture.py", line 880, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 167, in _multicall
  |     teardown.throw(outcome._exception)
  |     ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\skipping.py", line 257, in pytest_runtest_call
  |     return (yield)
  |             ^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\runner.py", line 174, in pytest_runtest_call
  |     item.runtest()
  |     ~~~~~~~~~~~~^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 1627, in runtest
  |     self.ihook.pytest_pyfunc_call(pyfuncitem=self)
  |     ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_hooks.py", line 513, in __call__
  |     return self._hookexec(self.name, self._hookimpls.copy(), kwargs, firstresult)
  |            ~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_manager.py", line 120, in _hookexec
  |     return self._inner_hookexec(hook_name, methods, kwargs, firstresult)
  |            ~~~~~~~~~~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 139, in _multicall
  |     raise exception.with_traceback(exception.__traceback__)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\pluggy\_callers.py", line 103, in _multicall
  |     res = hook_impl.function(*args)
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\_pytest\python.py", line 159, in pytest_pyfunc_call
  |     result = testfunction(**testargs)
  |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 22, in test_water_needed_no_crash
  |     def test_water_needed_no_crash(s):
  |                ^^^
  |   File "C:\Users\acous\AppData\Local\Programs\Python\Python313\Lib\site-packages\hypothesis\core.py", line 2334, in wrapped_test
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
```

```
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 24, in test_water_needed_no_crash
    |     water_needed(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 11, in water_needed
    |     need = float(plot) * 5 - float(rain)
    |            ~~~~~~~~~~~^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 26, in test_water_needed_no_crash
    |     assert False, f"water_needed crashed with: {e!r} for input {s!r}"
    | AssertionError: water_needed crashed with: ValueError("could not convert string to float: ''") for input '|'
    | assert False
    | Failing test case: test_water_needed_no_crash(
    |     s='|',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 24, in test_water_needed_no_crash
    |     water_needed(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 10, in water_needed
    |     plot, rain = text.split('|')
    |     ^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 26, in test_water_needed_no_crash
    |     assert False, f"water_needed crashed with: {e!r} for input {s!r}"
    | AssertionError: water_needed crashed with: ValueError('not enough values to unpack (expected 2, got 1)') for input ''
    | assert False
    | Failing test case: test_water_needed_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 24, in test_water_needed_no_crash
    |     water_needed(s)
    |     ~~~~~~~~~~~~~~^^^
    |   File "C:\college_files\se lab\lab 7\irrigation_scheduler.py", line 10, in water_needed
    |     plot, rain = text.split('|')
    |                  ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\college_files\se lab\lab 7\test_irrigation_scheduler.py", line 26, in test_water_needed_no_crash
    |     assert False, f"water_needed crashed with: {e!r} for input {s!r}"
    | AssertionError: water_needed crashed with: AttributeError("'NoneType' object has no attribute 'split'") for input None
    | assert False
    | Failing test case: test_water_needed_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
=================================== short test summary info ====================================
FAILED test_irrigation_scheduler.py::test_parse_zone_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_irrigation_scheduler.py::test_moisture_average_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_irrigation_scheduler.py::test_water_needed_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
==================================== 3 failed in 4.25s =====================================
PS C:\college_files\se lab\lab 7>
```

Step 5:

```
PS C:\college_files\se lab\lab 7> python irrigation_scheduler.py
==== Running parse_zone ====
Enter zone:area in m2 (e.g. north:120): north
Zone and Area: Invalid input

==== Running moisture_average ====
Enter moisture readings as CSV (e.g. 40,35,50): 40,,50
Average Moisture: Invalid input

==== Running water_needed ====
Enter area|rainfall (e.g. 120|8): 12
Water Needed: Invalid input
```

```
PS C:\college_files\se lab\lab 7> python irrigation_scheduler.py
 ==== Running parse_zone ====
 Enter zone:area in m2 (e.g. north:120): north:120
 Zone and Area: ('North', 120.0)

 ==== Running moisture_average ====
 Enter moisture readings as CSV (e.g. 40,35,50): 40,35,50
 Average Moisture: 41

 ==== Running water_needed ====
 Enter area|rainfall (e.g. 120|8): 120|8
 Water Needed: 592.0
PS C:\college_files\se lab\lab 7>
```

Step 6:

```
PS C:\college_files\se lab\lab 7> pytest test_irrigation_scheduler.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.0, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\acous\AppData\Local\Programs\Python\Python313\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\college_files\se lab\lab 7
plugins: anyio-4.15.1, hypothesis-6.170.15, langsmith-0.14.14
collected 3 items

test_irrigation_scheduler.py::test_parse_zone_no_crash PASSED                                                                      [ 33%]
test_irrigation_scheduler.py::test_moisture_average_no_crash PASSED                                                                [ 66%]
test_irrigation_scheduler.py::test_water_needed_no_crash PASSED                                                                    [100%]

============================================================ 3 passed in 0.62s ============================================================
PS C:\college_files\se lab\lab 7>
```